

RMO(INDIA) – 2001

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let BE and CF be the altitudes of an acute triangle ABC, with E on AC and F on AB. Let O be the point of intersection of BE and CF. Take any line KL through O with K on AB and L on AC. Suppose M and N are located on BE and CF respectively, such that KM is perpendicular to BE and LN is perpendicular to CF. Prove that FM is parallel to EN.
 2. Find all pairs p, q such that $p^2 + 7pq + q^2$ is the square of integers.
 3. Find the number of positive integers x , satisfying the condition: $\left[\frac{x}{99} \right] = \left[\frac{x}{101} \right]$, where $[x]$ denotes the greatest integer less than equal to x .
 4. Consider an $n \times n$ array of numbers:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix}$$

Suppose each row consists of the n numbers $1, 2, 3, \dots, n$ in some order and $a_{ij} = a_{ji}$ for $i = 1, 2, 3, \dots, n$ and $j = 1, 2, 3, \dots, n$. If n is odd, prove that the numbers $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ are $1, 2, 3, \dots, n$ in some order.

5. In a triangle ABC, D is a point on BC such that AD is the internal bisector of $\angle A$. Suppose $\angle B = 2\angle C$ and $CD = AB$. Prove that $m\angle A = 72^\circ$.
6. If x, y, z are sides of a triangle then prove that $|x^2(y - z) + y^2(z - x) + z^2(x - y)| < xyz$.
7. Prove that the product of the first 1000 positive even integers differs from the product of the first 1000 odd integers by a multiple of 2001.

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